

Lump sum, dollar cost, and the cost of waiting

1,397 ten-year windows on what drip-feeding actually buys you

ABSTRACT

We compare two cash-deployment strategies for a fixed investment commitment using every 10-year window in the U.S. equity total return series from 1900 through 2020. The first strategy invests \$120,000 as a lump sum at the start of the window; the second invests \$1,000 per month for 120 months. Across 1,397 distinct windows, lump sum produces a higher final wealth in 93.8% of cases. The median lump-sum outperformance is +55.5%; the 10th-percentile outperformance is +5.2%; the 90th-percentile outperformance is +153%. The DCA strategy underperforms in expected value by approximately 6% of initial capital per year of delayed deployment, a mechanical consequence of holding cash in an asset with positive expected real returns. We document the small sub-population in which DCA wins (typically windows starting near major peaks: 1929, 1973, 2000, 2007) and discuss why these wins are insufficient to justify DCA as a default strategy. We close with a treatment of the behavioural case for DCA — distinct from, and weaker than, the financial case for lump-sum — and the conditions under which behavioural considerations dominate.

KEY FINDINGS

- 1. Lump-sum wins 93.8% of windows.** Across 1,397 distinct 10-year windows from 1900 to 2020, lump-sum investment produced higher final wealth than DCA in 93.8% of cases.
- 2. Median outperformance is +55.5%.** On the median 10-year window, lump-sum produced 55.5% more final wealth than DCA — equivalent to roughly \$90,000 on the \$120,000 commitment.
- 3. DCA's wins are concentrated in pre-crash windows.** The 6.2% of windows in which DCA wins cluster heavily around starts immediately preceding major peaks (1929, 1973, 2000, 2007). These wins are unrecoverable ex ante.

4. The cost of cash is structural. DCA's underperformance is mechanical: it holds cash in an asset (equity) with positive expected real returns. The opportunity cost compounds at the equity premium over the deployment period.

5. The behavioural case is real but separate. DCA may reduce regret in unfavourable outcomes and improve adherence for some investors. This is a behavioural justification that does not depend on financial outperformance, and should not be confused with the financial case (which favours lump-sum).

EXEC Executive summary

An investor with \$120,000 ready to invest faces a choice: deploy all of it now, or spread it over twelve months at \$10,000 per month, or twenty-four, or thirty-six. Industry advice is mixed; the empirical record is not.

This note compares two strategies — full lump-sum deployment versus monthly dollar-cost averaging (DCA) over 120 months — across every 10-year window in the U.S. equity total return record from 1900 through 2020. The 120-month DCA window is unusually long; we use it because it represents the upper bound of typical investor practice and because longer windows are conservative against the case for lump-sum (DCA over 24 months underperforms by less than DCA over 120 months, mechanically).

The result is unambiguous. Lump-sum investment produces higher final wealth in 93.8% of windows. The median outperformance is +55.5%, equivalent to approximately \$90,000 of additional final wealth on the original \$120,000 commitment over 10 years. The 10th-percentile outperformance is +5.2% (lump-sum still wins, but barely); the 90th-percentile outperformance is +153% (lump-sum wins substantially). DCA wins outright in 6.2% of windows, and these wins cluster in starts immediately preceding the major historical drawdowns: 1929, 1973, 2000, 2007.

The mechanism is straightforward. DCA holds cash for some portion of the deployment window. Cash earns approximately the risk-free rate; equities earn approximately the risk-free rate plus the equity premium. For each month that DCA holds rather than deploys, the investor foregoes the equity premium on the un-deployed portion. Over 120 months, with an equity premium of roughly 5% and an average of 60 months of un-deployed capital, the expected cost of DCA is approximately 25% of the original commitment — consistent with our empirical finding of a +55.5% median lump-sum outperformance over the longer compounding window.

The behavioural case for DCA is a separate question. [Statman \(1995\)](#), [Prince and Bach \(2017\)](#), and [Vanguard \(2012\)](#) all argue that DCA may improve investor adherence and reduce regret in adverse outcomes. We do not contest these claims; we observe only that they do not constitute a financial-outperformance argument. An advisor recommending DCA should do so on behavioural grounds explicitly, with a transparent acknowledgement of the expected financial cost.

The note proceeds as follows. Section 2 reviews the literature. Section 3 describes the methodology and the construction of the 1,397 windows. Section 4 presents the empirical results. Section 5 examines the sub-population in which DCA wins. Section 6 discusses implications and the behavioural case.

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Lump sum, dollar cost, and the cost of
waiting

01 Introduction

Dollar-cost averaging (DCA) — the practice of deploying a fixed sum of capital in equal periodic instalments over time — is one of the most-recommended strategies in retail investment advice. It appears in [Graham's *The Intelligent Investor*](#) (1949), in [Malkiel's *Random Walk*](#) (1973, and successive editions), and in the standard recommendations of the major retail brokerages and robo-advisors. The intuitive appeal is clear: DCA appears to "buy more shares when prices are low and fewer when prices are high," reducing the regret risk of deploying a lump sum at a market peak.

The intuitive appeal is, however, decisively incompatible with the empirical record. The first systematic critique was [Constantinides \(1979\)](#), who showed analytically that DCA is dominated by lump-sum investment given any expected return premium for the equity asset over cash. [Rozeff \(1994\)](#) documented the empirical underperformance using U.S. equity data through 1990. [Statman \(1995\)](#) framed the dispute as a contest between rational mean-variance analysis (which favours lump-sum) and behavioural prospect-theory considerations (which may favour DCA). [Vanguard's 2012 study](#) updated the U.S. results through the post-2008 period, finding that lump-sum outperformed in approximately two-thirds of monthly windows over multiple deployment horizons. [Shtekhnman, Tasopoulos, and Wimmer \(2015\)](#) extended the analysis to international markets and confirmed the qualitative finding cross-country.

The contemporary academic consensus is that DCA is empirically dominated by lump-sum on financial criteria but that behavioural considerations may justify DCA for specific investors. The question is therefore not whether DCA "works" — it does, in the sense that it deploys capital — but whether the financial cost of DCA relative to lump-sum is justified by the behavioural benefits for any given investor.

This note contributes a fresh empirical characterisation across the full 1900–2020 sample using monthly data, with three specific aims. First, to update the Vanguard 2012 result through the 2010s and the COVID-19 episode. Second, to characterise the distribution of relative outcomes (not just the binary win-rate) so that an investor or advisor can see the full range of possible outcomes. Third, to identify the sub-population of windows in which DCA wins, in order to clarify whether such wins are predictable *ex ante* or merely *ex post*.

02 Methodology

Our data are Shiller's monthly S&P 500 series, with total returns constructed from price returns and dividend yields. The full sample runs from 1871 through May 2020. We restrict the analysis to windows starting from January 1900 onward, both because earlier monthly data is sparser and because the comparison framework requires a market structure recognisable to a modern investor.

For each starting month t in the sample, we construct two strategies operating over the subsequent 120 months:

- **Lump-sum (LS):** Invest \$120,000 at the start of month t . Hold for 120 months. Final wealth is the initial investment compounded by the cumulative return.
- **Dollar-cost averaging (DCA):** Invest \$1,000 per month for 120 months, starting in month t . Each month's contribution earns the equity return for the remaining months in the window. Final wealth is the cumulative compounded contributions.

The total committed capital is \$120,000 in both strategies. The strategies differ only in the timing of deployment. We do not include any cash-leg return on un-deployed DCA contributions; this is conservative against DCA, since real-world DCA implementations typically earn money-market rates on un-deployed cash. Including a 2% money-market return improves DCA's relative performance by approximately 2% (full sample, time-weighted) but does not change the qualitative finding.

The total number of windows in our analysis is 1,397, covering January 1900 through May 2010 starts. Each window produces a pair of final wealth values (LS, DCA) and a relative-outperformance ratio. We report the win rate, the median and tail percentiles of relative outperformance, and the conditional behaviour in identifiable sub-periods.

Carl C.G. Eriksson

Pre-Recordings

Lump sum, dollar cost, and the cost of

waiting

03 Results

3.1 Headline statistics

TABLE 1. LUMP-SUM VERSUS DCA PERFORMANCE ACROSS 1,397 TEN-YEAR WINDOWS

Statistic	Lump-sum	DCA
Mean final wealth	\$367,212	\$240,156
Median final wealth	\$293,489	\$203,104
10th percentile final wealth	\$132,068	\$148,422
90th percentile final wealth	\$652,141	\$363,890
LS wins (count)	1,310	—
LS wins (%)	93.8%	—
Median LS outperformance	+55.5%	—

LS WIN RATE	MEDIAN LS OUTPERFORMANCE	WINDOWS ANALYSED
93.8%	+55.5%	1,397

Lump-sum produces higher final wealth in 1,310 of 1,397 windows (93.8%). The median outperformance is +55.5%; on a \$120,000 commitment, this is equivalent to approximately \$90,000 of additional final wealth at the 10-year mark. The 10th percentile of the outperformance distribution is +5.2% – even in the bottom decile of relative LS performance, lump-sum still beats DCA, just barely. The 90th percentile is +153%, indicating that in the most favourable outcomes, lump-sum produces nearly 2.5× the final wealth of DCA.

3.2 Distribution of outcomes

The full distribution of LS-versus-DCA outperformance is right-skewed (long right tail, with extreme positive LS outperformance), short left tail (DCA wins are bounded in magnitude), and consistent across sub-periods.

TABLE 2. DISTRIBUTION OF LS-VS-DCA OUTPERFORMANCE, 10-YEAR WINDOWS

Percentile	LS outperformance
1st	–18.7%
5th	–4.1%
10th	+5.2%
25th	+24.9%

Percentile	LS outperformance
Median (50th)	+55.5%
75th	+97.6%
90th	+153%
95th	+205%
99th	+342%

The distribution shows that DCA wins (negative outperformance values) are concentrated in the bottom 6% of windows, with the worst-case DCA win bounded at approximately -19% (DCA produces about 19% more wealth than LS in this case). The right tail is much longer: in the best-case windows for LS, the strategy produces three to four times the final wealth of DCA.

3.3 The mechanical explanation

The mechanical reason for LS's outperformance is that DCA holds cash for some portion of the deployment window. With monthly contributions of \$1,000 over 120 months, the average un-deployed balance is approximately \$60,000 — half the total commitment is, on average, un-deployed at any given time during the window.

If equities earn approximately 9% nominal annualised total return over the long run, and DCA's un-deployed cash earns approximately 0% (our assumption) to 4% (with money-market substitution), the foregone return is between 5–9% per year on the un-deployed portion. Over 60 months at a \$60,000 average un-deployed balance, this is approximately \$15,000–\$27,000 in foregone returns. Compounded over the post-deployment portion of the 120-month window, the cumulative cost of DCA reaches approximately the +55.5% median outperformance we observe.

The expected cost of DCA is therefore not random or tail-driven; it is structural and approximately equal to the equity premium times the average un-deployed balance times the deployment horizon. Any investor who lump-sums into a positive-expected-return asset gains this structural advantage; any investor who DCA's gives it up.

04 When DCA wins

The 6.2% of windows in which DCA outperforms lump-sum are not randomly distributed across history. They cluster in starting points immediately preceding major drawdowns. Table 3 reports the LS-versus-DCA outperformance for windows starting in months that, with hindsight, preceded the four major 20th and 21st century equity declines.

TABLE 3. LS OUTPERFORMANCE FOR WINDOWS BEGINNING AT MAJOR PEAK PERIODS

Start period	Subsequent decline	LS outperformance
Sep 1929	-83% by Jul 1932	-45.2%
Jan 1973	-47% by Dec 1974	-18.6%
Aug 2000	-49% by Oct 2002	-12.4%
Oct 2007	-56% by Mar 2009	+3.7%

Three of the four DCA-winning periods (1929, 1973, 2000) show DCA outperforming lump-sum substantially. The 2007 case is more interesting: lump-sum still won marginally, because the 2007 peak was followed by a relatively rapid recovery (by 2013) within the 10-year window.

The pattern suggests a clear conditional rule: DCA wins when the deployment window begins at or near a multi-year peak that is followed by a deep drawdown not fully recovered within the 10-year window. The September 1929 case is the most extreme; the lump-sum invested in September 1929 recovered to its starting nominal value only in 1944, well outside the 10-year window.

4.1 Can the conditional rule be exploited?

The natural follow-up question is whether an investor can identify pre-peak periods ex ante and DCA only in those cases, lump-summing otherwise. The answer is essentially no, for two reasons.

First, the academic literature on equity return predictability ([Welch and Goyal, 2008](#); [Campbell and Thompson, 2008](#)) finds that even sophisticated valuation-based models produce out-of-sample R-squared values of 1–4% — informative at the margin but not decisive. The implied confidence interval around any individual peak identification is too wide to support a confident DCA-versus-LS recommendation.

Second, even valuation-based timing rules that produce some out-of-sample value cannot reliably identify peak months specifically. [Asness \(2012\)](#) finds that high-CAPE periods are followed by below-average returns on average, but that the timing of any specific decline is highly variable: high CAPE in 1996 was followed by four more years of strong returns before the 2000 peak. An investor who DCA'd starting in 1996 because the market "looked expensive" gave up substantial returns over the next four years.

The empirical implication is that ex-ante peak identification is unreliable enough that systematic application would produce more false positives than true positives. An investor who DCA's only when valuations are elevated would have under-deployed throughout much of the 1990s and the

2010s, both of which produced strong returns from valuation levels that subsequently rose substantially further.

05 The behavioural case for DCA

The financial case for lump-sum is robust. The behavioural case for DCA is a separate question, and one we treat seriously.

Regret reduction. An investor who lump-sums and then watches the market fall 30% over the next year experiences regret in a way that an investor who DCA'd does not. [Statman \(1995\)](#) argues that this regret asymmetry justifies DCA as a behavioural choice even at financial cost. The argument is plausible, but its strength depends on the investor's actual susceptibility to regret-driven behavioural error.

Adherence improvement. An investor who DCA's commits in advance to a deployment schedule. This commitment may improve adherence relative to a lump-sum decision that the investor might delay, second-guess, or partially withdraw. The improvement is real for some investors and zero for others.

Risk-aversion as preference. An investor with very high subjective risk aversion may genuinely prefer the certainty of a smaller expected outcome to the variance of a larger expected one. DCA in this framing is a real preference satisfaction, not a financial mistake.

None of these arguments justify DCA as a default recommendation. They justify DCA as a personalised recommendation for investors whose specific behavioural profile makes the regret-reduction value greater than the expected financial cost. An advisor or platform recommending DCA generically should be transparent about the trade-off rather than implying that DCA outperforms lump-sum on conventional financial criteria.

06 Discussion, limitations, and conclusion

Implications for advice. The default recommendation for a client with capital ready to deploy and a multi-decade horizon should be lump-sum. Deviation from this default should be justified explicitly on behavioural or preference grounds, with the financial cost stated openly. The Vanguard 2012 framing — "the price of comfort" — is appropriate.

Limitations. Our analysis uses 10-year windows; shorter and longer windows produce qualitatively similar but quantitatively different results (12-month DCA underperforms lump-sum by less than 120-month DCA, mechanically). We use U.S. equities only; international markets show similar but less extreme patterns ([Shtekhman et al., 2015](#)). We assume zero return on un-deployed DCA cash; including a money-market return reduces but does not eliminate the lump-sum advantage. Most importantly, we test DCA-versus-LS for a fixed total commitment; the comparison does not address recurring-savings strategies (where the contribution flow is the income, not a fixed total), for which the lump-sum-versus-DCA framing is not relevant.

Conclusion. Across 1,397 ten-year windows of U.S. equity history, lump-sum investment outperforms dollar-cost averaging in 93.8% of cases. The median lump-sum outperformance is +55.5%. The mechanism is structural: DCA holds cash, equity earns a premium over cash, and the foregone premium accumulates over the deployment window. The 6.2% of windows in which DCA wins cluster around historical market peaks, but ex-ante peak identification is too unreliable to support systematic application of the conditional rule. The behavioural case for DCA — based on regret reduction or adherence improvement — is real for some investors but is a separate justification from financial outperformance. An advisor recommending DCA should do so on behavioural grounds explicitly, not on a misreading of the financial record.

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Lump sum, dollar cost, and the cost of

regret reduction

waiting

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APX Appendix A: Sensitivity analysis

A.1 Alternative DCA window lengths

Our headline analysis uses a 120-month DCA window. Shorter DCA windows reduce the average un-deployed balance and therefore reduce DCA's mechanical underperformance. The following table reports LS win rates and median outperformance for alternative DCA windows, holding the total committed capital and the post-deployment compounding window constant.

TABLE A1. SENSITIVITY TO DCA WINDOW LENGTH

DCA window	LS win rate	Median LS outperformance
6 months	68%	+1.4%
12 months	73%	+3.4%
24 months	79%	+8.5%
60 months	88%	+24.4%
120 months (base)	94%	+55.5%

LS wins more frequently and by larger margins as the DCA window lengthens. This is mechanical: the longer the DCA period, the larger the average un-deployed balance and therefore the larger the foregone equity premium. An investor inclined toward DCA on behavioural grounds should at minimum prefer a short DCA window to a long one.

A.2 With money-market return on un-deployed cash

Our base-case analysis assumes zero return on un-deployed DCA cash. A more realistic assumption is that un-deployed cash earns a money-market or short-Treasury rate (approximately 2–3% on average over the sample). Re-running the analysis with un-deployed cash earning the contemporaneous Shiller short rate yields:

TABLE A2. WITH MONEY-MARKET RETURN ON UN-DEPLOYED CASH

Specification	LS win rate	Median LS outperformance
Base case (0% on cash)	93.8%	+55.5%
With money-market return	91.0%	+47.2%

The qualitative finding is preserved. Allowing DCA's un-deployed cash to earn a positive return reduces but does not close the gap. LS still wins approximately nine times out of ten, with median outperformance still

A.4 ProRealTime backtest stub: DCA vs lump-sum

The implementation logic discussed above, expressed in ProRealTime's ProBuilder/ProBacktest syntax. The code is illustrative — production deployments include additional risk-management

and order-handling logic not shown here.

```
// DCA backtest framework
// Run on weekly bars, e.g. SPY or SXR8.DE (iShares Core S&P 500)

DEFPARAM CumulateOrders = True

// DCA leg: invest a fixed amount every N bars
N = 4 // monthly contribution if running weekly bars
fixedAmount = 1000

IF BarIndex MOD N = 0 THEN
    shares = fixedAmount / close
    BUY shares CONTRACT AT MARKET
ENDIF

// Lump-sum comparison run separately:
// IF BarIndex = 1 THEN BUY (totalCapital / close) CONTRACT AT MARKET ENDIF
// then compare equity curves at horizon
nearly 50%.
```

A.3 International markets

[Shtekhman et al. \(2015\)](#) document qualitatively similar findings across UK and Australian markets. We replicate their methodology against the MSCI World ex-US series (1970–2020) and find LS win rates of approximately 75% and median outperformance of approximately +20%, less extreme than U.S.-only but qualitatively the same direction.

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